Pulse – Secure Wearable & Mobile Health Data Integration

Technical Design Documentation (v1.0)

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Contents

1 Overview

Pulse integrates wearable devices and mobile health platforms to securely collect physiological data and enhance personalized trigger detection. The platform continuously learns from user health patterns instead of relying only on manual symptom logging.

Collected metrics include:

- Heart Rate
- Resting Heart Rate
- Heart Rate Variability (HRV)
- Sleep Duration
- Stress Score
- Oxygen Saturation (SpO_2)
- Physical Activity
- Calories Burned
- Body Temperature (Supported Devices)

2 Objectives

- Secure health data collection
- User consent driven architecture
- Personalized trigger prediction
- Treatment effectiveness tracking
- Explainable AI recommendations
- Privacy-first implementation

3 Supported Platforms

Platform	Integration	Status
Android	Health Connect	Primary
Fitbit	Fitbit API	Primary
Apple Watch	HealthKit	Phase 2
Garmin	Garmin API	Phase 2
Oura Ring	Oura API	Phase 2
Samsung Health	Health Connect	Supported

4 System Architecture

```
Wearable Devices
      |
Official APIs
      |
OAuth 2.0
      |
FastAPI Backend
      |
Validation + Encryption
      |
Processing Layer
      |
Cognee Memory Graph
      |
Trigger Detection Engine
      |
Personalized Recommendations
```

5 Authentication

Pulse uses OAuth 2.0 authentication only.

1. User connects wearable.
2. Redirect to provider login.
3. User grants permissions.
4. Backend receives authorization code.
5. Access token stored securely.
6. Periodic synchronization begins.

6 Permission Management

Only minimum permissions are requested.

- Heart Rate
- Sleep
- Activity
- Oxygen Saturation

Never requested:

- Contacts
- Messages
- Photos
- Unnecessary Location

7 Synchronization Strategy

Synchronization occurs:

- Event driven where supported
- Scheduled every 30–60 minutes
- On application launch

8 Data Pipeline

```
Raw Sensor Data
      |
API Response
      |
Validation
      |
Unit Conversion
      |
Outlier Removal
      |
Aggregation
      |
Daily Summary
      |
Cognee Memory Graph
```

9 Stored Data Model

```
{
 "date":"2026-07-01",
 "average_heart_rate":81,
 "resting_heart_rate":63,
 "maximum_heart_rate":145,
 "sleep_duration":6.4,
 "steps":7420,
 "stress_score":72,
 "hrv":34,
 "spo2":98
}
```

10 Cognee Integration

Wearable metrics become graph entities:

- User
- Sleep
- Heart Rate
- Stress

- HRV
- Symptoms
- Medication
- Treatment

Relationships include:

- triggers
- alleviates
- correlates_with
- precedes

11 AI Pattern Detection

Example logic:

```
Sleep < 5 Hours
+
HRV Decrease
+
High Stress
      |
Historical Match
      |
Risk Prediction
      |
User Alert
```

12 Personalized Insights

Example recommendation:

> You slept only 4.8 hours. Similar conditions previously preceded headaches 4 out of the last 5 occurrences. Consider hydration and additional rest.

13 Security

Encryption

- TLS 1.3 in transit
- AES-256 at rest

Authentication

- OAuth 2.0
- JWT
- Refresh Tokens

API Security

- Role Based Access Control
- Audit Logs
- Rate Limiting
- Encrypted Token Storage

14 Privacy Principles

- Explicit consent
- Data minimization
- User-controlled deletion
- Revocable permissions
- No unauthorized sharing

15 Future Enhancements

- ECG Analysis
- Continuous Glucose Monitoring
- AQI and Weather Triggers
- Menstrual Cycle Tracking
- Doctor Dashboard
- Caregiver Portal
- Wellness Score
- Emergency Alerts

16 Recommended Tech Stack

Layer	Technology
Backend	FastAPI
Database	PostgreSQL + TimescaleDB
Memory Layer	Cognee
Cache	Redis
Scheduler	Celery/APScheduler
Wearables	Health Connect, Fitbit, HealthKit, Garmin, Oura
Monitoring	Prometheus + Grafana
Logging	ELK Stack

17 Conclusion

Pulse transforms from a symptom logging application into a proactive health intelligence platform by securely integrating wearable data, graph memory, and explainable AI to deliver personalized healthcare insights while maintaining strong privacy and security.